Department of Computer Science and Information Engineering

National Chung Cheng University

Programming (II) Final Project Proposal

A Simplified GTO-Inspired Texas Hold’em Decision System

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Chapter 1 Introduction

1-1 Research Motivation

Our team members are highly interested in Texas Hold'em. Texas Hold'em poker is a strategic card game that involves probability, uncertainty, and decision-making under incomplete information. In each hand, players must make decisions without knowing their opponents' hole cards or the future community cards. Therefore, making rational and optimized decisions is an important part of improving poker strategy.

Game Theory Optimal strategy provides a useful theoretical framework for balanced decision-making in poker. However, a complete GTO solver is highly complex and beyond the scope of our ability. For this reason, our project proposes a simplified GTO-inspired decision system for Texas Hold'em to design a practical and understandable algorithm that can support decision-making in simplified poker situations.

1-2 Problem Statement

The main problem of this project is how to evaluate a Texas Hold'em hand and recommend a reasonable action under uncertainty. Since the player does not know the opponents' hole cards or the future community cards, the program needs a method to estimate the hand's winning probability. In addition, a strong hand does not always mean that calling or raising is profitable, because the decision also depends on the pot size and the cost of continuing. Therefore, this project aims to develop a simplified GTO-inspired Texas Hold'em decision system that estimates hand equity through Monte Carlo simulation, calculates the expected value of possible actions, and recommends a reasonable decision based on the highest expected value.

Chapter 2 Implementation Method

2-1 Proposed Methodology

The proposed system will be implemented as a console-based program. The user will input the hero's hole cards, the number of opponents, the pot size, the call amount, the raise amount, and the number of simulations. The system will first create a standard 52-card deck and remove the hero's known cards. Then, it will repeatedly simulate random opponent hands and random community cards. For each simulation, the program will evaluate the best poker hand for the hero and each opponent. After many simulations, the system will estimate the hero's equity by calculating the proportion of wins and ties.

After estimating the equity, the system will calculate the expected value of different actions. The action with the highest expected value will be selected as the recommended decision.

In addition to the decision algorithm, the project will include a console-based user interface. The interface will use a menu-driven structure to guide the user through the decision analysis process. Users will be able to start a new analysis, view input format instructions, and exit the program. During the analysis, the system will ask for the hero's hole cards, number of opponents, pot size, call amount, raise amount, fold probability, and simulation count. After the calculation, the system will present the result in a clear format, including win rate, tie rate, loss rate, estimated equity, expected values, and the recommended action.

2-2 Algorithm Design

2-2-1 Monte Carlo Equity Estimation

The first main algorithm is Monte Carlo equity estimation. Since the unknown opponent cards and community cards cannot be predicted exactly, the system will randomly generate possible future outcomes many times. In each simulation, the program deals random cards to the opponents and random community cards on the board. The program then compares the hero’s hand with the opponents’ hands. The estimated equity is calculated as:

$$\text{Equity} = (\text{Wins} + 0.5 \times \text{Ties}) / \text{Total Simulations}$$

2-2-2 Poker Hand Evaluation

The second main algorithm is poker hand evaluation. In Texas Hold’em, each player uses two hole cards and five community cards to form the best possible five-card hand. The program will evaluate seven cards by checking hand categories from strongest to weakest, including straight flush, four of a kind, full house, flush, straight, three of a kind, two pair, one pair, and high card. The program will use rank counts and suit counts to identify hand categories.

2-2-3 Expected Value Decision Algorithm

The third main algorithm is expected value analysis. After estimating the equity, the system will calculate the expected value of possible actions. Folding has an expected value of zero in this simplified model. Calling is evaluated by comparing the cost of calling with the final pot size. Raising is evaluated using a simplified fold equity assumption.

EV_fold = 0

EV_call = Equity × Final Pot - Call Amount

EV_raise = Fold Probability × Current Pot + (1 - Fold Probability) × (Equity × Final Pot After Raise - Raise Amount)

The system will compare the expected values of fold, call, and raise, and recommend the action with the highest expected value.

2-3 Interface Design

The interface will be implemented as a console-based menu system. The main menu will provide three options: starting a new decision analysis, viewing the input format guide, and exiting the program. If the user starts a new analysis, the program will ask for the required information step by step.

The card input will use a rank-suit format. For example, Ah represents the ace of hearts, Ks represents the king of spades, Qd represents the queen of diamonds, and Tc represents the ten of clubs. This format allows the program to identify both the rank and suit of each card, which is necessary for evaluating flushes and straight flushes.

The interface will also include input validation. The program will reject duplicated cards, invalid ranks, invalid suits, negative pot sizes, invalid opponent numbers, and invalid simulation counts. After all inputs are accepted, the program will run the Monte Carlo simulation and expected value calculation. Finally, the results will be displayed in a structured output section.

2-4 Expected Features

Basic Features

1. Input hero's hole cards.
2. Input number of opponents.
3. Generate and shuffle a standard 52-card deck.
4. Randomly simulate opponent cards and community cards.
5. Evaluate poker hand rankings.
6. Estimate hand equity through Monte Carlo simulation.
7. Calculate EV for fold and call and raise.
8. Recommend a decision .
9. Display the result.

Advanced Features

1. Allow users to choose the number of simulations.
2. Design a more user-friendly hand input interface.
3. Display card input examples and validate illegal card combinations.
4. Allow users to input tournament-related information, such as stack sizes and remaining players.
5. Extend the decision model with advanced tournament poker concepts, such as Independent Chip Model (ICM), risk premium, and bubble factor.
6. Compare chip EV decisions with tournament EV considerations.
7. Provide explanations for how tournament pressure may affect the

recommended action.

8. Support different numbers of opponents.

Chapter 3 Timeline

Stage 1: Confirm the project topic and design the overall system structure.
Stage 2: Implement the basic card system, including card representation, deck generation, shuffling, and dealing.
Stage 3: Implement the poker hand evaluation algorithm to identify hand rankings such as pair, two pair, straight, flush, full house, and other poker hands.
Stage 4: Implement Monte Carlo simulation to estimate the winning probability of the hero's hand against one or more opponents.
Stage 5: Implement expected value calculation and decision recommendation for actions such as fold, call, and raise.
Stage 6: Design and implement the user interface, including menu options, input guidance, and input validation.
Stage 7: Develop selected advanced features, including simplified tournament poker concepts such as Independent Chip Model (ICM), risk premium, and bubble factor as adjustment factors.
Stage 8: Test the system, debug errors, improve the output explanation, and prepare the final project demo and final report.

By the end of this project, we expect to complete a console-based Texas Hold'em decision system. The system will be able to estimate the winning probability of a given hand through Monte Carlo simulation, calculate the expected value of fold, call, and raise, and recommend a simplified decision. The final program will demonstrate how probability, simulation, and expected value analysis can be applied to poker decision-making.

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